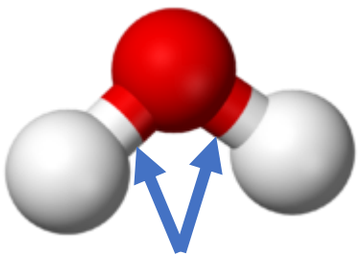
Knowledge Organiser
Year 7: Chemicals – Mixtures and Solutions

Mixtures vs. Compounds
- A **compound** is a molecule formed from two or more **different elements** chemically bonded together e.g. H_2O
- A mixture is made from two or more substances which have been **physically mixed**
- Compounds contain strong **chemical bonds** so it is hard to separate the elements
- The substances in a mixture are not chemically bonded so are **easy to separate**



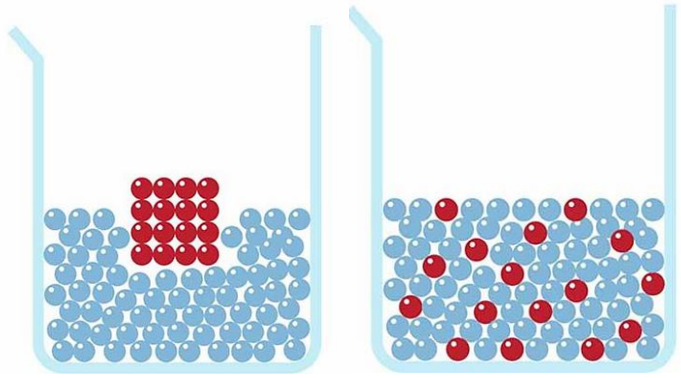
The components of a mixture can be separated without chemical reactions



Strong chemical bonds

The elements in a compound are chemically bonded and difficult to separate

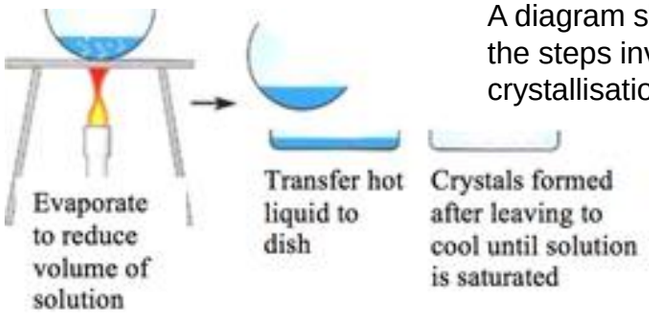
What is a solution?
- A solution is a **liquid mixture** where one part is **dissolved** in the other e.g. salt water
- A solute is dissolved in a solvent when it's particles are broken up into such small pieces that we can't see them anymore
- If a given solute **interacts strongly** with a solvent, it is said to be **soluble**
- If a substance cannot be dissolved easily we say it is **insoluble**



A particle diagram showing how the particles in a sugar cube change when it is dissolved in water

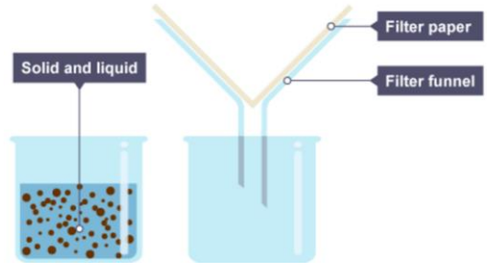
Keyword	Definition
Mixture	Two or more substances physically combined
Solution	A liquid mixture where one part is dissolved in the other
Solvent	The liquid which does the dissolving
Dissolve	The process of mixing a solute into a solvent to form a solution
Solute	The substance that dissolves
Soluble	When a substance can dissolve in a given solvent

Separating a solution: crystallisation
- A **soluble solid** can be separated from a solution by **crystallisation**
- The solution is heated to a high enough temperature that the **solvent** begins to boil
- Once solid crystals of the **solute** begin to appear, stop heating
- The **solution** can then be left to evaporate slowly over a few days
- Uniform **crystals** of the solute will crystallise from the solution

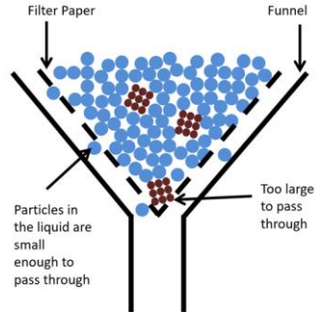


A diagram showing the steps involved in crystallisation

Separating a solution: filtration
- If an **insoluble solid** is mixed in a solvent it can be removed through filtration e.g. sand in water
- A filter works by having small holes
- These holes are large enough to allow the solvent molecules through e.g. water
- But small enough to prevent the solid particles from going through e.g. sand



A diagram showing the equipment used for filtration



A particle diagram which shows why filtration cannot separate sand from water

Why can't we filter salt from water?
- Salt water is a solution
- The salt is dissolved in the water
- This means the particles of salt are very small, so small we can no longer see them
- If we pour salt water through a filter, the salt particles will be small enough to fit through the holes, so will not be separated from the solution

RAG Yourself!

Learning Outcomes;	RAG
Compare chemical compounds to mixtures	
Explain why compounds are hard to separate	
Define the key terms: solution, solvent, solute and soluble	
Draw a particle diagram to show how a solution is formed	
Describe how to separate a soluble solid from a solution by evaporation	
Explain why filtration cannot separate salt and water	

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Mixtures vs. Compounds



The components of a mixture can be separated without chemical reactions

The elements in a compound are chemically bonded and difficult to separate

What is a solution?

A particle diagram showing how the particles in a sugar cube change when it is dissolved in water

Keyword	Definition
Mixture	
Solution	
Solvent	
Dissolve	
Solute	
Soluble	

Separating a solution: crystallisation

A diagram showing the steps involved in crystallisation

Separating a solution: filtration

A diagram showing the equipment used for filtration

A particle diagram which shows why filtration cannot separate sand from water

Why can't we filter salt from water?

RAG Yourself!

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